From Eliciture to Internal Representation: Implicit Causality in Language Models

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Abstract

Do language models merely behave as if they infer implicit causes, or do they encode causal relationships that generalize? Four studies test behavioral elicitation, relation selectivity, causal OOD generalization, and naturalistic transfer. Across six models, behavioral conclusions depend on elicitation method. On 150 causal OOD pairs, causal directions reach .97–1.00 accuracy, compared with .57–.95 for temporal and .41–.69 for plausibility directions. Demographic substitutions also shift causal projections in 65 matched scenarios per model. These studies connect pragmatic behavior to internal representation while exposing important construct-validity limits.

1 Introduction

Cohen and Kehler call defeasible pragmatic enrichments *conversational elicitures* [Cohen and Kehler, 2021]. Language models reproduce human-like eliciture judgments and relative-clause preferences [Pan and Kehler, 2025], but behavioral agreement does not reveal whether they encode the inferred relation. Because linear representations can reveal event-plausibility distinctions obscured by output probabilities [Lepori et al., 2026], we use contrastive activation addition (CAA) [Rimsky et al., 2024] to study implicit causality.

We ask three questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Study 1: Measuring Behavioral Eliciture

We vary the response measure in a conceptual replication of Pan and Kehler, finding substantial measurement dependence.

2.1 Methods

Models. We evaluate the base checkpoints of Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024].

Dataset and analysis. We evaluate 150 critical/control pairs with yes/no judgments, expected seven-point Likert scores, and the relative probability of the same continuation after *because* versus bare *and*. The last measure uses no metalinguistic prompt. A pair is correct when the critical item receives

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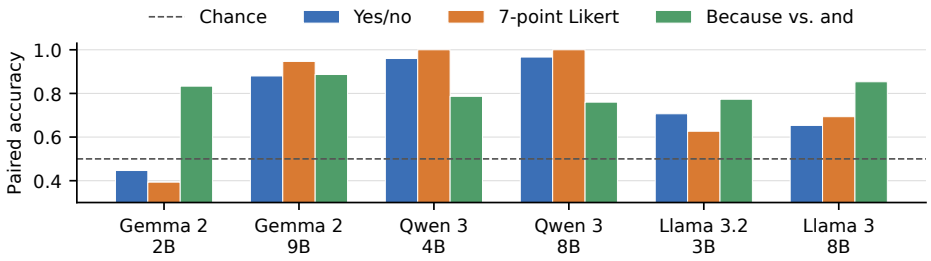


Figure 1: **RQ1.** Paired accuracy on 150 eliciture pairs. Dashed line: chance. Behavioral conclusions vary substantially with response mapping, particularly for smaller models.

the stronger causal response, allowing us to compare prompting with probability measurement [Hu and Levy, 2023].

2.2 Results

Figure 1 shows that Qwen3 4B/8B reach 1.00 with Likert responses and at least .96 with yes/no, but fall to .79/.76 under probability assignment. Conversely, Gemma 2 2B scores .45/.39 with prompting but .83 with *because* versus *and*. Overall, no interface supports a model-independent conclusion.

3 Study 2: Relation-Selective Causal Representations

We compare cross-sentence causal directions with temporal and plausibility directions, finding strong diagonals and weaker cross-transfer.

3.1 Methods

Datasets. Causal, temporal, and plausibility directions respectively use 384/96, 800/200, and 56/14 train/ID-development pairs.

Analysis. At every fourth layer and the final layer, CAA averages final-token differences between matched positive and negative texts. We select the earliest layer maximizing mean own-development paired accuracy and ROC–AUC, then apply that frozen direction to every target relation. Matrix diagonals measure within-relation discrimination; off-diagonals measure cross-transfer. This is an ID-development analysis, not an OOD test.

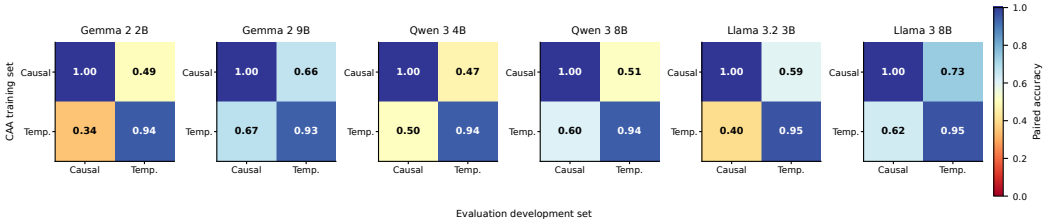


Figure 2: **RQ2a, causal vs. temporal.** ID-development paired accuracy when each row's CAA direction is evaluated on each column's relation. Each panel is one model; diagonals measure own-relation discrimination and off-diagonals measure cross-transfer.

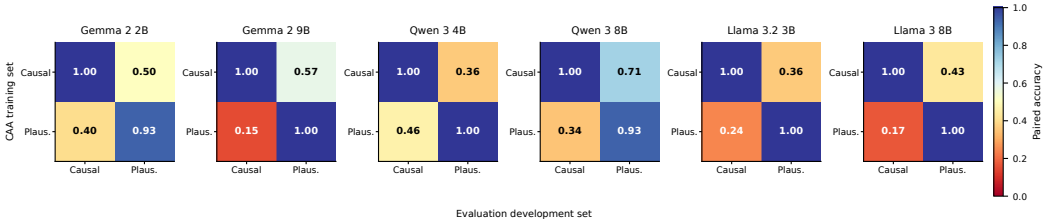


Figure 3: **RQ2a, causal vs. plausibility.** Six-model cross-transfer matrices; plausibility cells ($n = 14$ pairs) are exploratory.

3.2 Results

Figures 2–3 show causal self-transfer of 1.00 for all six models, temporal self-transfer of .93–.95, and plausibility self-transfer of .93–1.00. However, causal-to-temporal accuracy is .47–.73 and causal-to-plausibility is .36–.71; reverse transfer reaches .34–.67 and .15–.46. The gaps support relation selectivity, while occasional above-chance transfer argues against strict orthogonality.

4 Study 3: Generalization to Causal OOD

We next test whether causal directions remain selectively informative on unseen causal scenarios.

4.1 Methods

Datasets. Causal and temporal CAAs use 1,000/200 and 800/200 train/development pairs; the plausibility CAA uses 70 probable/impossible pairs with five-fold layer selection. We evaluate every frozen direction on the same 150-pair causal OOD set.

Analysis. We project each critical/control sentence onto the corresponding direction. A pair is correct when the critical item projects farther than its matched control.

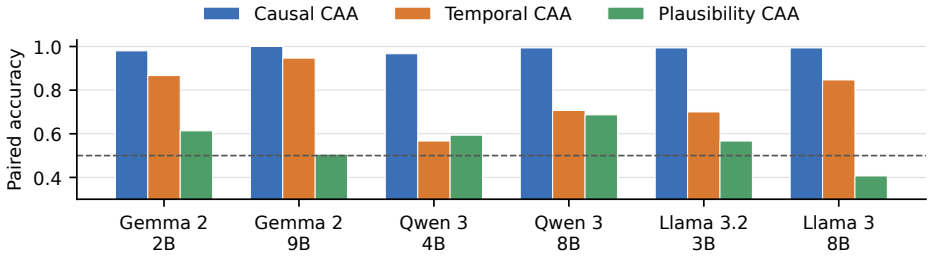


Figure 4: **RQ2b.** Paired accuracy of causal, temporal, and plausibility CAAs on 150 causal OOD pairs. Dashed line: chance.

4.2 Results

Figure 4 shows that causal CAA reaches .97–1.00 paired accuracy across models. Temporal CAA varies from .57–.95, whereas plausibility CAA ranges from .41–.69. The causal direction therefore generalizes most consistently to new causal scenarios; transfer from the comparison directions is weaker and model-dependent.

5 Study 4: Generalization to Naturalistic Scenarios

We finally test causal representations on matched naturalistic scenarios.

5.1 Methods

Dataset. We evaluate 65 self-blinding scenarios per model in matched White-male, marginalized-male, and marginalized-female versions.

Analysis. A unit causal direction fit separately on 1,000 within-sentence pairs projects each scenario's final-token state. We report within-scenario differences from the White-male version with 10,000-sample bootstrap 95% intervals and visualize raw, unnormalized CAA projections.

Table 1: **RQ3.** Mean paired unit-CAA change ± bootstrap 95% CI half-width (65 scenarios/model).

Model	Marginalized male − White male	Marginalized female − White male
Gemma 2 2B	0.41 ± 0.32	0.57 ± 0.32
Gemma 2 9B	1.9 ± 0.40	2.3 ± 0.41
Qwen3 4B	0.073 ± 0.093	0.15 ± 0.10
Qwen3 8B	0.29 ± 0.17	0.39 ± 0.15
Llama 3.2 3B	0.025 ± 0.029	0.069 ± 0.033
Llama 3 8B	0.030 ± 0.020	0.060 ± 0.021

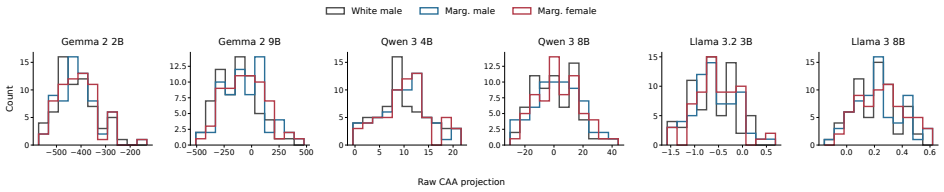


Figure 5: **RQ3.** Raw CAA projections by matched condition (65 scores/condition/model).

5.2 Results

Table 1 shows positive estimates for both contrasts in every model. Marginalized-female intervals exclude zero for all six; marginalized-male intervals do so for four. However, Figure 5 shows substantial distributional overlap. These associations do not establish social bias, behavior, or a causal mechanism.

6 Discussion

Behavioral evidence is measurement-sensitive, whereas causal directions are relation-selective and generalize to causal OOD data. However, Study 2 uses ID development sets (only 14 plausibility pairs), CAA is correlational, and 65 scenarios cannot establish general behavior. Larger tests, interventions, and behavioral validation remain necessary.

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